

JEE MAIN REVISION MASTER: INVERSE TRIGONOMETRIC FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	17 PYQs	Quantitative Analysis
Very High	Inverse Trigonometric Functions — Core Principles & Governing Laws	15 PYQs	Core Principle
Moderate	Inverse Trigonometric Functions — Applied Cases, Reactions & Traps	5 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	$\cos(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{33}{65}), \quad \text{quad}$ $f(x)=\tan^{-1}(\sin x-\cos x), [0,\pi], \quad \text{quad } 2-2x^2=1+2x^2$
Inverse Trigonometric Functions — Core Principles & Governing Laws	$f(x)=\sin^{-1}2x+\sin 2x+\cos^{-1}2x+\cos 2x, x \in [0,\frac{\pi}{8}], \quad \text{quad } S=\{x \in \mathbb{R}: 0<x<1 \text{ and } 2\tan^{-1}(\frac{1-x}{1+x})=\cos^{-1}(\frac{1-x^2}{1+x^2})\}, \quad \text{quad}$ $\cos^{-1}(x)-2\sin^{-1}(x)=\cos^{-1}(2x)$
Inverse Trigonometric Functions — Applied Cases, Reactions & Traps	$\frac{\pi}{2} \leq x \leq \frac{3\pi}{4}, \quad \text{quad } b=\cos^{-1}(\cos(5)), \quad \text{quad}$ $-\frac{1}{2\sqrt{6}}<x<\frac{1}{2\sqrt{6}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes (Very High Yield — 17 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If the domain of the function $f(x) = \sin^{-1}(\frac{1}{x^2-2x-2})$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta =$..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "If $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2})$, $x \in (0, 1)$, then the value of x is:..." [Ans: A]
- [2026 Evening Shift] Q#4 (MCQ): "Considering the principal values of inverse trigonometric functions, the value of the expression $\tan(2\sin^{-1}(\frac{2}{\sqrt{13}})) - 2\cos^{-1}(\dots)$ " [Ans: N/A]

3.2 Inverse Trigonometric Functions — Core Principles & Governing Laws (Very High Yield — 15 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Inverse Trigonometric Functions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Inverse Trigonometric Functions PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha = 3\sin^{-1}(\frac{6}{11})$ and $\beta = 3\cos^{-1}(\frac{4}{9})$, where inverse trigonometric functions take only the principal values..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "Let $0 < \alpha < 1, \beta = (1/3\alpha)$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = (\pi/4)$. Then $6(\alpha + \beta)$ is equal to:..." [Ans: B]
- [2026 Morning Shift] Q#3 (MCQ): "If the domain of the function $f(x) = \cos^{-1}(\frac{2x-5}{11-3x}) + \sin^{-1}(2x^2-3x+1)$ is the interval $[\alpha, \beta]$, then $\alpha + 2\beta =$..." [Ans: N/A]

3.3 Inverse Trigonometric Functions — Applied Cases, Reactions & Traps (Moderate Yield — 5 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2026 Morning Shift] Q#2 (MCQ):** "The number of solutions of $\tan^{-1}4x + \tan^{-1}6x = (\pi/6)$, where $-\frac{1}{\sqrt{6}} < x < \frac{1}{\sqrt{6}}$, is equal to ..." **[Ans: N/A]**
- **[2025 Morning Shift] Q#2 (MCQ):** "If $(\pi/2) \leq x \leq (3\pi/4)$, then $\cos^{-1}((12/13)\cos x + (5/13)\sin x)$ is equal to..." **[Ans: N/A]**
- **[2024 Evening Shift] Q#2 (MCQ):** "If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Let $\alpha = 3\sin^{-1}((6/11))$ and $\beta = 3\cos^{-1}((4/9))$, where inverse trigonometric functions take only the principal values. Given below are two statements : Statement I : $\cos(\alpha + \beta) > 0$. Statement II : $\cos(\alpha) < 0$. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: $(1/2) < (6/11) < \frac{1}{\sqrt{2}} \Rightarrow \sin^{-1}((1/2)) < \sin^{-1}((6/11)) < \sin^{-1}(\frac{1}{\sqrt{2}}) \dots (i) \Rightarrow 3\sin^{-1}((1/2)) < 3\sin^{-1}((6/11)) < 3\sin^{-1}(\frac{1}{\sqrt{2}}) \Rightarrow (\pi/2) < \alpha < (3\pi/4) \Rightarrow \cos \alpha < 0 \dots (ii) (4/9) < (1/2) < \frac{1}{\sqrt{2}} \Rightarrow \cos^{-1}((4/9)) > (\pi/3) > (\pi/4) \Rightarrow 3\cos^{-1}((4/9)) > \pi > (3\pi/4) \Rightarrow \beta > \pi \dots (ii)$ Inequalities (i) and (ii) shows the mathematical logic for $\alpha + \beta > \frac{\pi}{2}$

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Evening Shift (Q#1)

If the domain of the function $f(x) = \sin^{-1}(\frac{1}{\sqrt{1-x^2-2x-2}})$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

- (A) 4
- (B) 2
- (C) 5
- (D) 3

Correct Answer: None

Solution Analysis: Domain of $\sin^{-1}\theta$ is $-1 \leq \theta \leq 1$ $f(x) = \sin^{-1}(\frac{1}{\sqrt{1-x^2-2x-2}})$ is defined if $-1 \leq \frac{1}{\sqrt{1-x^2-2x-2}} \leq 1$ s.t $\frac{1}{\sqrt{1-x^2-2x-2}} \geq -1$ and $\frac{1}{\sqrt{1-x^2-2x-2}} \leq 1$ so these two inequality and final solution will be intersection of solution of these two inequality. solve $\frac{1}{\sqrt{1-x^2-2x-2}} \geq -1$ $\frac{1}{\sqrt{1-x^2-2x-2}} + 1 \geq 0$ $\frac{1+x^2-2x-2}{\sqrt{1-x^2-2x-2}} \geq 0$ $\frac{(x^2-2x-1)}{\sqrt{1-x^2-2x-2}} \geq 0$ roots of x^2-2x-1 is $x = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}$ roots of x^2-2x-2 is $x = \frac{2 \pm \sqrt{4+8}}{2}$

Benchmark Problem 3: JEE Main 2026 (Online) 22nd January Morning Shift (Q#2)

The number of solutions of $\tan^{-1}4x + \tan^{-1}6x = (\pi/6)$, where $-\frac{1}{\sqrt{6}} < x < \frac{1}{\sqrt{6}}$, is equal to :

- (A) 2
- (B) 0
- (C) 3
- (D) 1

Correct Answer: None

Solution Analysis: $\tan^{-1}4x + \tan^{-1}6x = (\pi/6)$ $\tan^{-1}(\frac{10x1-24x^2}{24x^2+10\sqrt{3}}) = (\pi/6)$ $24x^2+10\sqrt{3}x-1=0$ $x = \frac{-5\sqrt{3} \pm 3\sqrt{14}}{24}$